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Master Thesis

# The impact of tropospheric versus stratospheric $CO_2$ forcing on the Hadley cell

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# Abstract

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# Nomenclature

## Abbreviations

| Abbreviation | Definition |
|---|---|
| USA | United States of America |
| EU | European Union |

## Variable Names

| Symbol | Definition | Unit |
|---|---|---|
| $\alpha$ | greek alpha | [N] |
| $\beta$ | greek beta | [J] |

# 1 Introduction

The Hadley cell (HC) is the dominant meridional large-scale circulation in the tropics. Its ascending branch at the equator is collocated with inter-tropical convergence zone (ITCZ), which is characterized by large amounts of precipitation. On the other hand, the dry, semi-arid and arid regions of subsidence in the subtropics are associated with the descending branch of the HC. The equatorward flowing air in the lower branch is deflected westward by the Coriolis force, forming the so called trade wind. Equivalently, the anti-trades are formed by the poleward flowing air of the upper branch, which deflected eastward. In the annual mean and in the equinoxes there are two similar overturning cells in the tropics - one in each hemisphere. It extends from 0 to 30 °N in the Northern hemisphere and equivalently 0 to 30 °S in the Southern hemisphere. Throughout the annual cycle there are strong variations in the HC's extent and strength: As the maximum of insolation moves poleward, so does the ITCZ and the HCs, resulting in a strong cross-equatorial winter (hemisphere without the maximum) and a weaker summer HC (hemisphere with the maximum). This seasonal migration is related to the global monsoon, which has been identified as an angular momentum conserving HC regime (Bordoni & Schneider, 2008).

The HC is responsible for the poleward transport of energy and the equatorward transport of moisture, thereby reducing the meridional temperature gradients imposed by solar insolation. The HC has strong implications for human and animal life as well as agriculture and vegetation on this planet. Its extent crucially modulates global precipitation patterns by setting the regions of ascend and descend. The strength of the HC determines what amount of energy is reaching the off-equatorial region and what amount of moisture is reaching the equator, setting the potential precipitation amounts. This is particularly evident in the Sahel region, which is located approximately 10 °N of the equator on the African continent. This semi-arid region receives most of its precipitation in the boreal summer when the HC and the ITCZ migrate northward. Thus, the rainfall in the Sahel is dynamically linked to the HC position. Changes in the HC also affect the livelihood of the local population and economies that remain heavily dependent on agricultural production.
This has also been shown for paleoclimate. Paleoclimate simulations of the HC display large changes in HC strength and extend during the last glacial maximum and the mid holocene due to changes in orbital forcing (D'Agostino et al., 2017). This is also supported by evidence from tropical paleoclimate proxy records of precipitation changes (McGee et al., 2014). This past evidence of strong HC changes combined with the large societal impact of

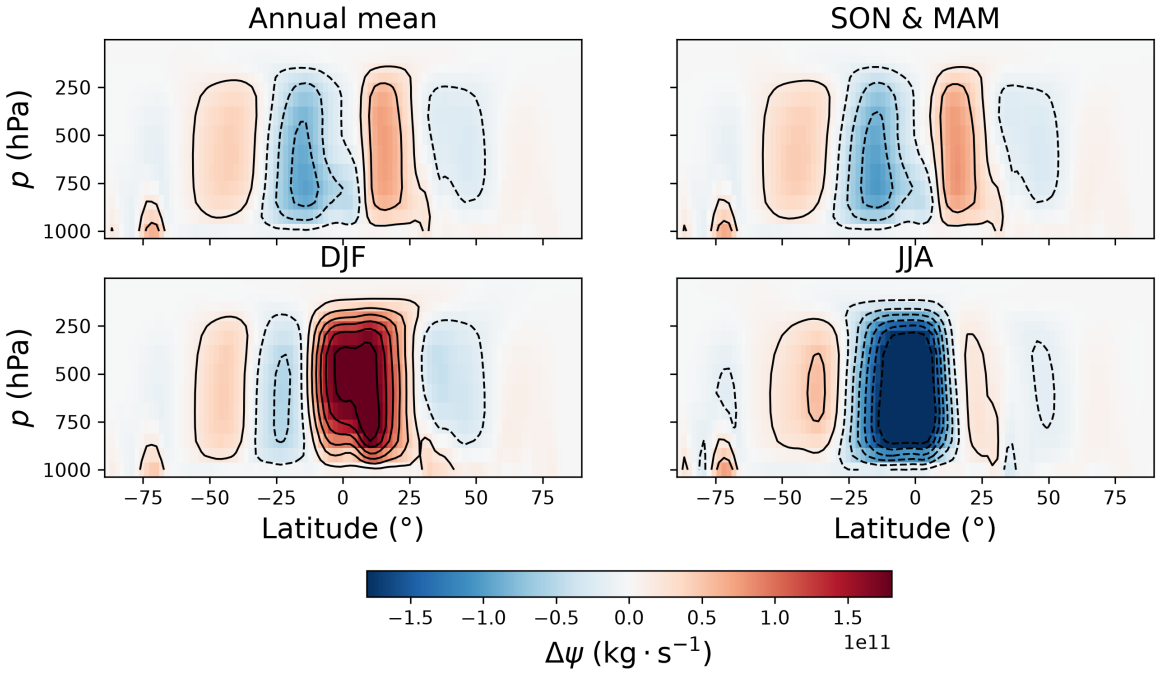



**Figure 1.1:** TODO: use correct PI control simulations and add annual mean while removing double equinox

precipitation patterns give a strong reason to investigate the projected HC changes of human driven greenhouse gas emissions in the current era.

The $CO_2$ in the atmosphere is very uniformly distributed. The forcing that it imposes is not. The forcing is critically set by the difference between surface and stratospheric temperature, which varies strongest in the meridional direction. Given that the HC is flattening the meridional energy gradient, the response of the HC strength depends on meridional forcing structure. This has been shown for quadrupling of $CO_2$ only in certain latitude belts in coupled simulations (Kim et al., 2022). A forcing in mid latitudes weakens the Hadley circulation whereas a forcing at the equator leads to it strengthening. We investigate how the *vertical* structure of CO2 forcing, rather than its horizontal structure, alters atmospheric circulation. We achieve this by running simulations with a General Circulation Model (GCM) while quadrupling $CO_2$ only in the troposphere and only in the stratosphere using a pre-industrial control tropopause height to separate the two cases. To our knowledge, these experiments have not yet been described by the literature.

Climate simulations from the Coupled Model Intercomparison Project (CMIP), have established a HC weakening and widening under climate change (Lu et al., 2007; Xia et al., 2020). Explaining this, previous studies have emphasized the importance of drivers related to the troposphere (Chemke & Polvani, 2021; Fan & Dommenget, 2024). However, historical simulations show that stratospheric forcing also influences the general circulation. In particular stratospheric ozone depletion has led to a cooling of the Southern hemisphere lower-stratospheric polar cap. This has been identified as the main driver for the widening of the Hadley circulation over the second half of the twentieth century (Polvani et al., 2011). The goal of our study is to disentangle the contributions of tropospheric and stratospheric $CO_2$ to the established HC weakening with uniform $CO_2$ distribution. We find that the HC response

to tropospheric $CO_2$ is a strengthening and widening in both hemispheres. This opposes our expectations evoked from the literature, which identifies tropospheric warming parameters as the main drivers for HC weakening under uniform $CO_2$ concentrations. Our results suggest that the weakening of the HC originates from stratospheric cooling caused by stratospheric $CO_2$. To investigate and explain the different responses of the HC strength we employ angular momentum and energy budget frameworks. We use the axisymmetric angular momentum framework to attribute HC changes to changes in tropopause height, meridional temperature, gradient, and vertical stability. Since this theory ignores the contribution of eddies to HC strength we also investigate changes in eddy divergence to explain the observed changes in the mean circulation. The energetic framework is a diagnostically unifying approach. It uses changes in top-of-atmosphere (TOA) radiative fluxes and moist static energy to distinguish different contributing factors to HC changes.

We contribute to the advancement of the field by revising the applicability of different HC frameworks and simplified models. Splitting up $CO_2$ forcing in two vertical layers puts existing theories to the test and gives the opportunity to rethink accepted truths in climate science. Overall, this is a contribution to foundational research that advances our physical understanding of the drivers of general circulation changes.

TBD **Overview over the study** Is this really necessary?

# 2

# Theory

TODO: Describe general structure of the chapter

## 2.1. CO2 forcing in the atmosphere and its temperature response

$CO_2$ is relatively well mixed in the atmosphere. In our simulations with a general circulation model though, we can prescribe different vertical $CO_2$ profiles. Since $CO_2$ is a greenhouse gas, this changes the radiative balance and thus the temperature profile of the atmosphere. This section is describing the temperature response of the atmosphere to a uniform increase in $CO_2$ concentration and to provide an expectation for the temperature change in the different vertical $CO_2$ profiles. In particular we care about the tropopause height, and meridional and vertical temperature gradients, which are important to understand changes in the HC strength and extent as we will show in Section **??**.

### 2.1.1. Tropospheric warming and stratospheric cooling

The temperature response to a uniform increase in $CO_2$ concentration is characterized by a warming in the troposphere and a cooling in the stratosphere. The vertical temperature profile of the atmosphere can be determined as a first approximation from the budget of incoming and outgoing radiation everywhere. This state is called radiative equilibrium (RE). Solving RE with a numerical transfer code for a non-grey atmosphere, which means including the vertical distribution of radiatively active gases, yields a lapse rate larger than observed in the atmosphere (Manabe & Strickler, 1964). The observed lapse rate on Earth is close to $6.5\,\mathrm{K\,km^{-1}}$. It can be shown that the RE temperature profile is highly sensitive to vertical perturbations, so it is convectively unstable. To account for the upward motions and the condensational heating (moist and dry convection), the concept of radiative convective equilibrium (RCE) was introduced. Here, we adjust the lapse rate to the critical lapse rate (stability threshold), whenever it is exceeded in the integrations of the radiative transfer code. This model was famously extended by assuming fixed relative humidity (RH) within the troposphere (Manabe & Wetherald, 1967). This is a good approximation because the absolute humidity in the atmosphere strongly depends on temperature, so does the saturation water pressure (CC). This model is very successful in explaining the atmosphere's response to increased greenhouse gas emissions. It provided the first realistic estimate of climate sensitivity.

The temperature in the stratosphere is governed by the relation of heating due to the absorption of shortwave radiation by ozone and the emission of longwave radiation due to $CO_2$. One of the most robust responses of increasing the $CO_2$ concentration in the atmosphere has been the cooling of the stratosphere (Santer et al., 1996). A qualitative explanation goes like this: Because the heating due to ozone stays constant but the emissivity of $CO_2$ is increased, the atmosphere has to cool locally to stay in radiative balance. Since the stratosphere adjust faster, the stratospheric cooling also occurs faster than the warming in the troposphere. Near the tropopause changes in stratospheric temperature are almost negligible, while they increase with height until they reach their maximum approximately at the stratopause. Here, each doubling of $CO_2$ decreases the temperature by around 8 K. More insights into the structure of stratospheric cooling can be obtained from a spectroscopic analysis of $CO_2$ forcing in the atmosphere (Cohen et al., 2026).

An increase in tropopause height due to a uniform increase of $CO_2$ is robust at all latitudes in most models. A simple model for a grey atmosphere predicts that the HC changes either due to changes in surface temperature or in lapse rate. Increasing the surface temperature and decreasing the lapse rate pushes the tropopause up (Vallis et al., 2015). This is what happens under uniform $CO_2$ forcing. We would expect the same response for a tropospheric $CO_2$ increase. For a stratospheric $CO_2$ increase, we would expect no response in the tropopause height, since surface temperature and the lapse rate are not affected by stratospheric cooling.

Separating the $CO_2$ concentration change into a tropospheric and a stratospheric component will have a very different response in terms of the temperature anomaly field. For an increase of tropospheric $CO_2$ concentration, we expect a warming signal from the troposphere due to the increased absorption of $CO_2$. For an increase of stratospheric $CO_2$ concentration, we expect a cooling signal from the stratosphere due to the increased emission.

### 2.1.2. Warming structure and feedbacks

The warming and cooling pattern in the atmosphere due to a $CO_2$ increase does not yet reveal how vertical and meridional temperature gradients change. In the tropics the warming aloft is amplified (top heavy). This is due to the increased moisture content which reduces the lapse rate. This constitutes a negative feedback. Due to the greenhouse gases a substantial fraction of OLR originates in the mid troposphere, where now the warming is amplified, which increases the OLR (Stefan-Boltzmann law). Another important paradigm to explain the horizontal profiles of temperature within the tropics is the weak temperature gradient approximation (Sobel et al., 2001). It argues that the Coriolis force is too weak in the tropics to sustain horizontal gradients of pressure or temperature. Diabatic heat fluxes are communicated throughout the tropics by gravity waves maintaining the horizontal homogeneity. These two concepts combined provide a good framework to understand the warming structure with an of $CO_2$ concentration in the tropical troposphere: The vertical temperature gradient is reduced which means that the atmosphere is more stable. Vertical temperature gradients within the tropics are expected to remain equal due to to the weak temperature gradient approximation.

In the Arctic, we have the somewhat opposite pattern. Warming is amplified at the surface (bottom heavy), and generally larger than at the rest of the globe. This phenomenon is called Arctic amplification. Enhanced warming is neither yet expected nor observed in the Antarctic, where it is delayed by enhanced ocean heat uptake (Armour et al., 2016) and the

large elevation of Antarctic continental land mass (Hahn et al., 2020). Warming in the Arctic is enhanced during the winter months not during the summer as one would expect from the ice albedo feedback. The enhanced sea ice melt during summer increases the OLR and heat fluxes from the newly opened waters, warming the atmosphere (Dai et al., 2019). This local effect is enhanced by remote influence on the local lapse rate. Climatologically, there is a surface-based inversion because advective moist heating keeps the atmosphere warm compared to the surface which strongly cools to space. The warming associated with the sea ice loss weakens the inversion, producing a positive lower lapse rate feedback. Generally, the atmosphere in the Arctic is quite dry and due to inversion convection cannot transport moisture into the free troposphere. This diminishes the upper lapse rate feedback in the Arctic decreasing the OLR relative to lower latitudes, further warming the Arctic (Feldl et al., 2020). The enhanced warming at lower level in the Arctic compared to the global mean, increases the meridional near-surface temperate gradient.

## 2.2. Historical and projected Hadley cell trends

The HC response to a uniform increase in $CO_2$ concentration and the associated temperature response is very well described in the literature. HC changes have been observed already since the beginning of the satellite era. Observational datasets using outgoing longwave radiation, precipitation, and sea level pressure as proxies for the HC position have shown a significant widening of the HC in the historical period (1979-2000s) (Hu et al., 2011). Depending on the observed quantity the widening has a magnitude of 1.2 to 4.0°. Similar trends have also been found in reanalysis data (Hu & Fu, 2007). In phase 6 of the Coupled Model Intercomparison Project (CMIP6) "All-forcing" experiments also found that the Hadley cell widened in the period 1970-2014 by 0.65° ± 0.02° per decade. TODO This is much smaller than the trends obtained from observational records. The influence of natural variability is highlighted in this context. The widening in CMIP6 models is found to be larger in the Southern hemisphere. The primary driver of the historical widening is greenhouse gases forcing. Changes in anthropogenic aerosol forcing and stratospheric ozone only contribute to Southern hemisphere HC changes (Xia et al., 2020).

In terms of the HC strength, studies found opposing trends in climate models and reanalysis data in the historical period for the Northern hemisphere. While CMIP5 models show a robust weakening of the NH HC, the reanalysis data even suggests a strengthening. This discrepancy is not due to internal variability but rather stems from a misrepresentation of latent heating in the reanalysis (Chemke & Polvani, 2019). According to CMIP6 models, greenhouse gas forcing leads to a weakening in both hemispheres, with stratospheric ozone, and aerosol forcing contributing to a net modest strengthening in the southern hemisphere. This strengthening trend had not been observed in the previous CMIP5 models. The annual mean Northern hemisphere HC weakened by approximately 5% over 1970-2014 (Xia et al., 2020).

CMIP6 projections of the HC extent show that the widening trend of the HC increases with radiative forcing in both hemispheres. This increase is stronger in the annual mean Southern hemisphere HC while showing no seasonality. The strength of the SH HC is not projected to change with radiative forcing. In the NH, there is a projected net weakening of the annual-mean maximum of about 10% over the period of 2016-2100 (Xia et al., 2020).

Consistent with the historical simulation the largest contributor to the change is expected to be the increase in future $CO_2$ concentrations in the atmosphere.

## 2.3. Dynamical theories of the Hadley circulation

Given the temperature responses to a tropospheric, or stratospheric $CO_2$ forcing, we want to discuss their implications on the Hadley circulation. Thus, we investigate what HC changes are to be expected according to different HC frameworks. The conservation of angular momentum sets the physical concept to understand the movements of fluids, the circulation of air, in the atmosphere imposed by a thermal structure. In the following subsections, we will introduce two different dynamical theories of the HC and discuss their respective predictions for HC strength and extent changes.

### 2.3.1. Axisymmetric theory

Trying to answer the question, why there is a meridional circulation in the first place Held and Hou (1980) developed the axisymmetric theory of the HC. This question is raised because one could imagine a world where the Earth is just in radiative equilibrium (see Section **??**) everywhere. This means that every point on Earth emits just as much radiation as it receives from the sun. Thus, there would not be any meridional heat flux necessary. The axisymmetric theory considers the zonal angular momentum equation in the $\phi$–$p$ plane with no meridional circulation ($v = \omega = 0$). Being zonally symmetric prohibits the existence of large scale barotropic or baroclinic eddies. Choosing a radiative-equilibrium temperature profile, which resembles that of the Earth,

$$\frac{\theta_E}{\theta_0} = 1 - \frac{1}{3}\Delta_h(3\sin^2\phi - 1) + \Delta_v\left(\frac{z}{H} - \frac{1}{2}\right) \tag{2.1}$$

(see Figure 2.1a), one can obtain a corresponding zonal wind $u_E$ via the thermal wind relationship. Here, $\Delta_h$ is the fractional pole to equator temperature difference, $\Delta_v$ the fractional temperature difference between the surface and the top of the fluid, $H$ the depth of the fluid, and $\theta_0$ the global mean value of $\theta_E$. This solution not only produces westerly winds at the equator, which are not observed in the atmosphere, but also violate Hide's theorem (Hide, 1969). Hide's theorem states that a maximum of angular momentum within the flow domain cannot be sustained due to dissipation within the flow. A maximum of angular momentum can only exist at the boundaries. This tells us that this RE temperature profile cannot persist on Earth and that we require some poleward transport of heat to flatten the meridional temperature gradient.

The conservation of angular momentum

$$M = (u + \Omega a\cos\phi)a\cos\phi \tag{2.2}$$

provides a second constraint for a HC solution with the zonal wind $u$, $a$ being the radius and $\Omega$ the rotational frequency of the Earth. The angular momentum conserving wind can then be derived as

$$u_M = \Omega a\frac{\sin^2\phi}{\cos\phi}. \tag{2.3}$$

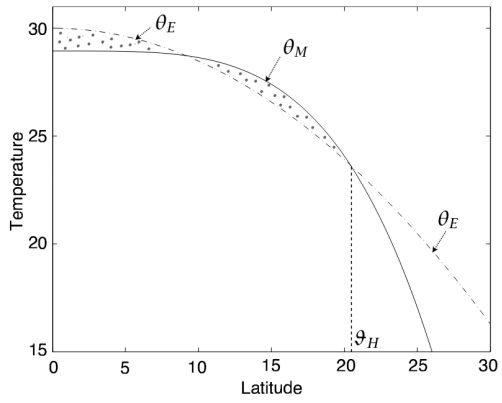
This allows us to infer, via the thermal wind relationship, a temperature profile $\theta_M$ (see Figure 2.1a), which is now slaved to angular momentum conservation. Assuming that the energy is conserved in the upper branch and that the solution is continuous at the edge of the HC, allows us to determine the undefined parameters of this solution, one of which is the latitude of the HC edge

$$\phi_H \sim \left( \frac{gH\Delta_h}{\Omega^2 a^2} \right)^{1/2}. \tag{2.4}$$

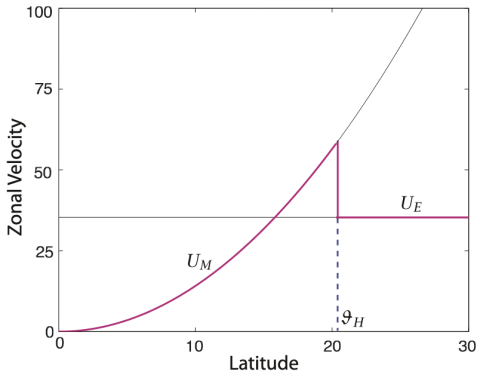
From the temperature equation we also obtain a scaling for its strength

$$\psi_{\max} \sim \frac{1}{\tau} \frac{g^{3/2}(H\Delta_h)^{5/2}}{\Delta_v a^2 \Omega^3}. \tag{2.5}$$

Here, $g$ is the gravitational acceleration of Earth and $\tau$ is the time scale at which $\theta$ relaxes to $\theta_E$.




**(a)** The radiative-equilibrium temperature profile $\theta_E$ is flattened to the angular-momentum-conserving temperature $\theta_M$ by the HC.




**(b)** The angular-momentum-conserving zonal wind $u_M$ increases with latitude. Poleward of the HC edge, the zonal wind from the radiative-equilibrium solution $u_E$ is valid.

**Figure 2.1:** Latitudinal profiles of temperature and zonal wind under radiative equilibrium and under angular momentum conservation of the axisymmetric model, from Vallis (2017).

The key contribution of this theory is to explain why there is a HC with finite width in the first place. It is to reduce the meridional temperature gradients imposed by solar insolation.

For a uniform increase of $CO_2$ we know that the tropopause height increases, which would lead to a HC widening and strengthening according to Equations (2.4) and (2.5). The vertical temperature gradient of potential temperature increases (more stable atmosphere), which would lead to a weakening of the HC according to Equation (2.5). The meridional temperature gradient is expected to decrease at lower levels in the atmosphere due to arctic amplification while it is expected to increase aloft due to stratospheric cooling on the poleward flank of the subtropical jet and the tropospheric warming at the equatorward flank. Thus, we cannot use the axisymmetric theory to qualitatively predict the HC response to a uniform increase of $CO_2$ in the atmosphere.

For an increase of tropospheric $CO_2$, we expect a warming of the troposphere, which enhanced aloft and amplified in the arctic. This will increase the tropopause height, decrease

the vertical temperature gradient, and decrease the meridional temperature gradient at lower levels while increasing it aloft. This reflects the same situation as for a uniform increase of $CO_2$, which is why we expect a weakening and a widening of the HC (as is evident in the uniform case). Just from the scaling laws of (2.4) and (2.5) we cannot predict the qualitative change of the HC with an increase of tropospheric $CO_2$.

For an increase of stratospheric $CO_2$, we expect a cooling of the stratosphere. In our axisymmetric framework this only affects the meridional temperature gradient, which is increased at the location of the subtropical jet. From this we would expect a modest widening and strengthening from the axisymmetric scalings if any effect at all.

The axisymmetric theory predicts that the HC depends directly on thermal driving via the scaling laws. Subsequent studies have advanced this theory also to temperature profiles which have their maximum off the equator, in order to account for seasonal variations (Lindzen & Hou, 1988). In the axisymmetric model, the HC is weaker in equinox conditions than observed, leading to the suggestion that the HC in the annual mean is not close to equinox conditions but rather that the HC flips between solstitial states. More recent studies have addressed this discrepancy using eddy permitting models (Walker & Schneider, 2005). In these models, the strength of an annually averaged Hadley circulation is comparable to the Hadley circulation driven by an annually averaged heating. This means that the HC in its equinox state resembles very much an annual mean HC. This highlights the importance of eddies in driving the HC strength, which will be covered in the next section.

### 2.3.2. Eddy driven mean circulation

The importance of eddies in shaping the HC was acknowledged even before the axisymmetric theory was developed (Kuo, 1956). More recent studies have compared axisymmetric models to eddy permitting simulations to determine the importance of eddies in altering HC strength (Singh et al., 2017; Walker & Schneider, 2006). They have found that the eddies increase the HC strength by 50–75%, depending on the model.

From the eddy perspective the HC edge is set by the latitude at which the zonal winds become baroclinically unstable. Baroclinic instability arises due to large vertical shear in a flow. This is the source of eddies. Using a two layer model, one can derive the HC edge as (Held, 2000)

$$\phi_H \sim \left(\frac{NH_e}{\Omega a}\right)^{1/2} \tag{2.6}$$

with $N$ being the Brunt-Väisälä frequency and $H_e$ the depth of the troposphere in the extratropical regions. It is important to note that this is a different scaling than the one derived from the axisymmetric theory. The eddy perspective predicts that the HC edge is set by extratropical parameters, whereas the axisymmetric theory predicts that it is set by tropical parameters.

To quantify HC strength changes from the eddy perspective, we apply a Reynolds decomposition on the zonal and meridional wind. We obtain the eddies as deviations in time from the zonal and temporal mean flow

$$u = \bar{u} + u', \quad v = \bar{v} + v'. \tag{2.7}$$

These deviations are associated with weather and Rossby waves in the mid-latitudes. Inserting this into the zonal momentum equation and neglecting viscous forces and vertical advection,

we obtain

$$f\bar{v}(1 - Ro) = \frac{1}{a\cos^2\phi}\frac{\partial(\cos^2\phi\ \overline{u'v'})}{\partial\phi} + \frac{\partial\overline{u'\omega'}}{\partial p}. \tag{2.8}$$

Here, the Rossby number is

$$Ro = \frac{\bar{\xi}}{f} \tag{2.9}$$

where $\xi = \hat{k} \cdot \nabla \times \vec{u}$ is the relative vorticity. Equation (2.8) has two limits. For $Ro \to 1$, the left-hand side becomes zero and the eddy term becomes negligible. This means that the Hadley circulation is close to angular momentum conservation

$$\frac{\partial\overline{M}}{\partial\phi} = 0 \tag{2.10}$$

as described by the axisymmetric solution in Section 2.3.1. For $Ro \to 0$, the meridional wind $\bar{v}$ is instead balanced by the eddy momentum flux convergence. This regime can be understood as eddy-driven (see Figure 2.2). For the intermediate $Ro$, no complete dynamical theory exists (Schneider, 2006).

In the eddy driven limit, the climatological divergence of eddy momentum flux, which is observed in the tropics, requires a positive mean meridional wind. This holds even if the zones of baroclinic instability lie well poleward of the HC edge because Rossby waves can propagate into the HC, producing a poleward momentum flux. At the HC edge there is $\bar{v} = 0$ and thus no eddy momentum flux. The eddy momentum flux divergence also explains the equatorward meridional wind in the eddy-driven Ferrel cell in mid-latitudes. Over the seasonal cycle, the Rossby number varies between 0.2 and 0.8, approaching either of the two limits. During the equinoxes the Rossby number is small which indicates a strong eddy influence. During the solstices the Rossby number is closer to 1 in the winter HC but still small in the summer HC.

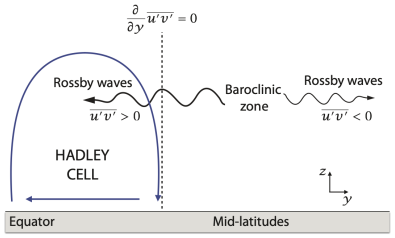



**Figure 2.2:** The eddy driven HC, from (Vallis, 2017)

Assuming that the HC is entirely eddy driven ($Ro = 0$), the HC strength can be predicted from the eddy momentum flux by inserting Equation (2.8) into Equation (3.1). Following Walker and Schneider (2006), this gives the integrated eddy momentum flux divergence

$$\psi_{\max} = \frac{2\pi a\cos\phi}{fg}\int_{p_{\max}}^{0}\left(\frac{1}{a\cos^2\phi}\frac{\partial(\cos^2\phi\ \overline{u'v'})}{\partial\phi} + \frac{\partial\overline{u'\omega'}}{\partial p}\right)\mathrm{d}p' \equiv \mathcal{T} \tag{2.11}$$

with $p_{\max}$ being the pressure level at which the MSF becomes maximum and the Coriolis parameter $f$. The quantity $\mathcal{T}$ is a predictor for the HC strength given a certain eddy momentum flux divergence. The HC is thus only indirectly coupled to thermodynamic drivers, which set the zones of baroclinic instability. It has been shown to be skillfully predict the HC strength in eddy permitting models for various climates (Walker & Schneider, 2006).

From the eddy framework, one can predict how the temperature response to $CO_2$ influences the HC. Therefore, it is crucial to look at how the zones of baroclinic instability change. This is set by the changes in meridional temperature gradient via the thermal wind relationship. The decrease in meridional temperature gradient at the surface opposes the effect of an increase in meridional temperature gradient aloft in both the uniform and the tropospheric $CO_2$ increase case. Which effect dominates the jet change, is still discussed in the literature (Ossó et al., 2024).

## 2.4. The energy budget framework of the Hadley circulation

The energy budget analysis provides a diagnostic perspective on the HC. It cannot explain why the Earth is not in radiative (convective) equilibrium, but provides a framework for understanding its energy transport mechanisms. Thus, it circumvents the Rossby number regime dependency, providing a more universal approach. Important to notice is that this framework can only deliver a prognostic relationship between energy transport and HC strength. Changes in HC extent cannot be investigated.

Using TOA radiative fluxes (see Figure 2.3), one can infer that the short-wave radiative flux at the equator is much larger than the emitted outgoing long wave radiation. The opposite is true for the poles. This implies a poleward energy transport by the atmosphere and the ocean. For every column of the earth system, the energy tendency can be written as (Hartmann,

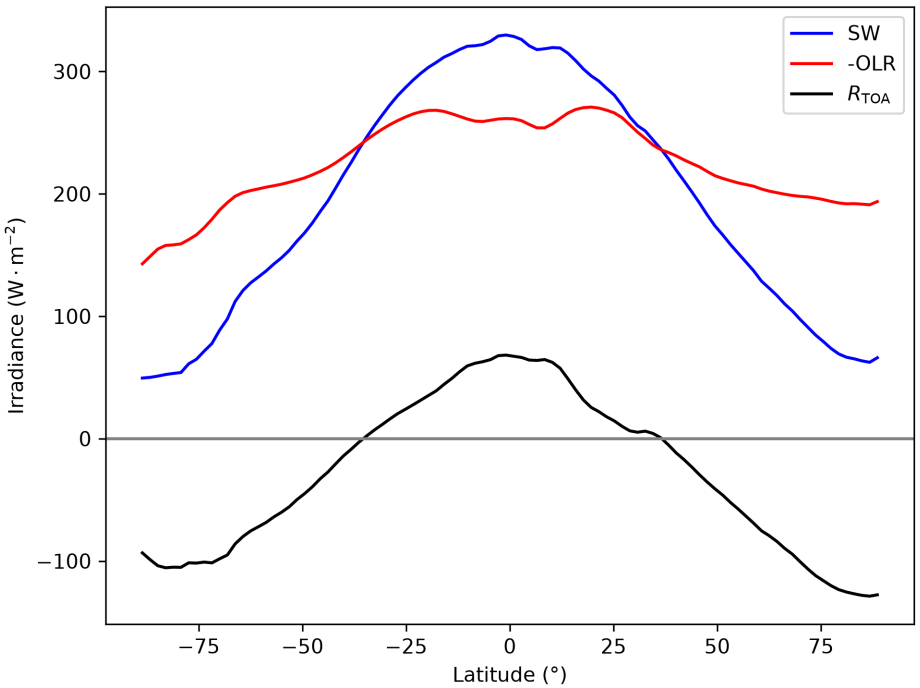



**Figure 2.3:** Zonal and annual mean average of irradiance from PI control simulations.

2016; Oort & Haar, 1976)

$$\frac{\partial E_{\text{ao}}}{\partial t} = R_{\text{TOA}} - \Delta F_{\text{ao}} \tag{2.12}$$

where $R_{\mathrm{TOA}}$ is the net radiative flux at the top of the atmosphere, which consists of the net shortwave radiation plus the net longwave radiation (positive values indicate a downward flux), and $\Delta F_{\mathrm{ao}}$ is the divergence of energy transport by the atmosphere and ocean.

Integrating this equation in the steady state over a latitude band gives

$$F_{ao}(\phi) = \int_{-\pi/2}^{\phi} \int_{0}^{2\pi} R'_{\mathrm{TOA}} a^2 \cos\phi \mathrm{d}\lambda \mathrm{d}\phi' \tag{2.13}$$

which provides a measure for the energy transport at every latitude. We use the anomalous net radiative TOA flux $R'_{\mathrm{TOA}}$ here, subtracting the global mean value, to make sure that the energy transport is closed (zero transport at both poles) (Zelinka & Hartmann, 2012). To separate between the energy transport by the atmosphere $F_a$ and the ocean, one can subtract the surface energy fluxes, which can be obtained from the sum of surface latent, sensible heat, and net shortwave and longwave fluxes, from the TOA radiative fluxes.

To understand the changes in the energy transport are related to changes in the HC, we need to quantify the energy transport by the HC. For this, we need to determine the energy within an air parcel in the atmosphere, which is given by the moist static energy (MSE). It is defined as

$$h = c_p T + gz + L_v q \tag{2.14}$$

with $c_p$ being the specific heat capacity at constant pressure, $g$ the gravitational acceleration, $z$ the height, $L_v$ the latent heat of vaporization and $q$ the specific humidity. The transport of MSE by the mean meridional circulation at a certain latitude $\phi$ is given by vertical integration of the meridional wind $\bar{v}$ multiplied by the MSE $h$ over the entire atmospheric column

$$F_{\mathrm{HC}}(\phi) = \frac{2\pi a \cos\phi}{g} \int_{0}^{p_s} [\bar{v}][\bar{h}] \mathrm{d}p. \tag{2.15}$$

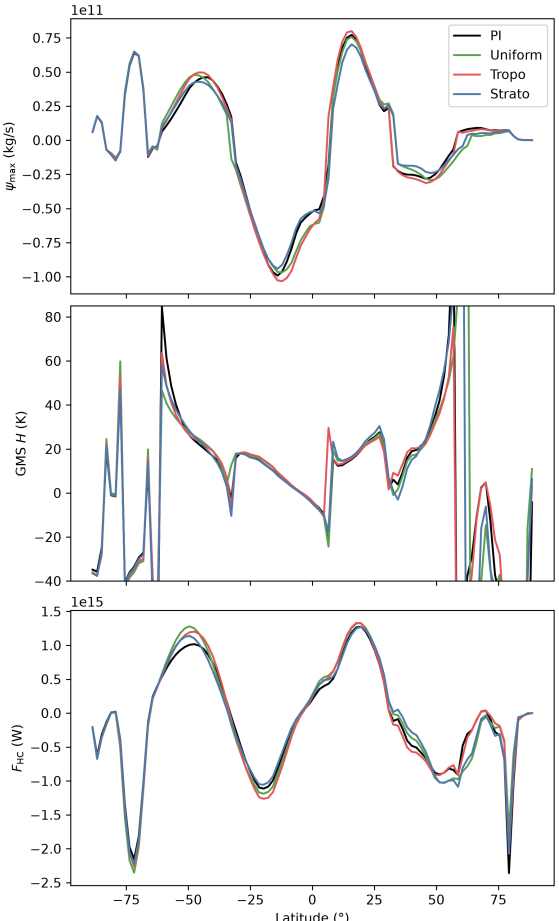
This expression compares the poleward energy transport by the upper branch of the HC to the equatorward energy transport by the lower branch. Expressing the HC strength $\psi_{\mathrm{max}}$ in terms of the energy transport $F_{\mathrm{HC}}$ and the gross moist stability $H$ gives

$$F_{\mathrm{HC}} = \psi_{\mathrm{max}} H \tag{2.16}$$

Gross moist stability (GMS) is defined as the ratio of energy transport to mass transport by the HC which can be understood as the energy transport efficiency. It is inferred to

$$H(\phi) = \frac{2\pi a \cos\phi}{g} \frac{1}{\psi_{\mathrm{max}}} \int_{0}^{p_s} [\bar{v}][\bar{h}] \mathrm{d}p. \tag{2.17}$$

It is a measure for the stratification of the atmosphere. A large GMS indicates a strongly stratified atmosphere. To understand the changes



in HC strength, we linearize Equation (2.16) and obtain

$$\frac{\delta\psi_{\max}}{\psi_{\max}} = \frac{\delta F_{\mathrm{HC}}}{F_{\mathrm{HC}}} - \frac{\delta H}{H}. \tag{2.18}$$

This yields the contribution of GMS and HC transport changes. Knowing the total energy transport in the atmosphere $F_{\mathrm{a}}$ from Equation (3.2), it can be further split up into a mean $F_{\mathrm{HC}}$ and an eddy energy flux component $F_{\mathrm{e}}$ (see Figure 2.5), which gives

$$F_{\mathrm{a}} = F_{\mathrm{HC}} + F_{\mathrm{e}}. \tag{2.19}$$

The eddy component can be calculated as the difference between $F_{\mathrm{a}}$ and $F_{\mathrm{HC}}$ or by using the transient eddy fluxes of MSE

$$F_{\mathrm{e}}(\phi) = \frac{2\pi a \cos\phi}{g} \int_0^{p_s} [\overline{v'h'}]\mathrm{d}p'. \tag{2.20}$$

Inserting this into the linearized Equation (2.16) gives

$$\frac{\delta\psi_{\max}}{\psi_{\max}} = \frac{\delta F_{\mathrm{a}}}{F_{\mathrm{HC}}} - \frac{\delta F_{\mathrm{e}}}{F_{\mathrm{HC}}} - \frac{\delta H}{H}.$$

The HC changes are now a function of the total change in energy transport. The changes in $R_{\mathrm{TOA}}$ can be further split up into a change in radiative forcing $R_f$ and a change due to feedbacks $\sum_i \lambda_i \Delta T_s$, which gives

$$\Delta R_{\mathrm{TOA}} = \sum_i \lambda_i \Delta T_s + R_f. \tag{2.21}$$

The forcing is the temperature-independent effect of CO2 on the TOA radiative fluxes and the feedbacks are the changes in TOA radiative fluxes due to changes in surface temperature. This change in surface temperature has an effect on the surface albedo, water vapor, clouds, and lapse rate, and OLR. An effect that with increasing surface temperature increases or decreases the downward radiation is called a positive or negative feedback, respectively.

Inserting Equation (2.21) into Equation (3.2) gives us the final expression for the HC strength changes decomposed into the contributions of forcing $\delta F_{\mathrm{forc}}$, feedbacks $\delta F_{\mathrm{fdb}}$, eddies $\delta F_{\mathrm{e}}$, and GMS $\delta H$

$$\frac{\delta\psi_{\max}}{\psi_{\max}} = \frac{\delta F_{\mathrm{forc}}}{F_{\mathrm{HC}}} + \frac{\delta F_{\mathrm{fdb}}}{F_{\mathrm{HC}}} - \frac{\delta F_{\mathrm{e}}}{F_{\mathrm{HC}}} - \frac{\delta H}{H}. \tag{2.22}$$

GMS is expected to increase with global warming, which is according to this framework contributing to a weakening of the HC. Again, using the known changes of these parameters under uniform $CO_2$ forcing, we are trying to qualitatively predict their contribution to the HC changes in our respective forcing cases. A uniform increase of $CO_2$ increases the GMS, which contributes to a weakening of the HC. An increase in eddy heat transport also contributes to a decreases in HC strength, because more of the necessary transport carried out by the

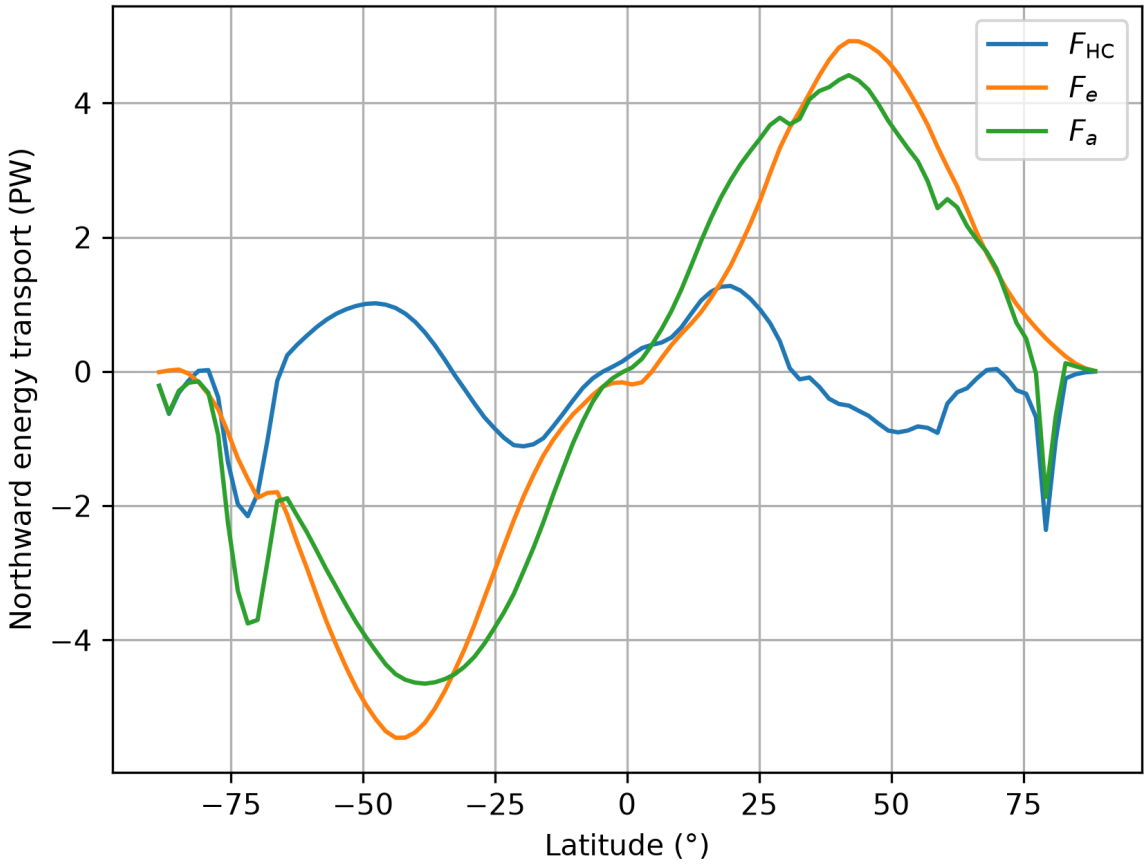



**Figure 2.5:** Northward energy transport

eddies. TODO: In our particular setup (see Section **??**), there are no changes in the HC due to a change in ocean heat uptake (OHU). In CMIP models, studies find a decrease in OHU at the equator relative to the extra-tropics, which is responsible for a strengthening of the HC (Feldl & Bordoni, 2016). Due to the meridional forcing structure, which is set by the surface temperature (Jeevanjee et al., 2021) and therefore enhanced in the tropics relative to the extra-tropics, one finds a positive forcing contribution. Using the radiative kernel technique allows us to estimate the influence of the individual climate feedbacks on the HC changes. The contribution of a feedback is determined by its meridional structure not by it being positive or negative, because it is always evaluated against its meridional mean. A feedback that is larger or smaller than its mean value in the tropics compared to the extra-topics or the poles strengthens or weakens the HC, respectively. For a uniform increase in $CO_2$ concentration this is the case for the water vapor feedback. The lapse rate, Planck and surface albedo feedback contribute to a weakening of the HC. The cloud feedback shows a large model dependence and can either contribute positively or negatively (Feldl & Bordoni, 2016).

Transferring this to the tropospheric $CO_2$ case almost all aspects seem identical here. There only might be some uncertainty again regarding the eddy component. This means the HC response should be similar as for a uniform increase of $CO_2$,i.e., a weakening of the HC.

For an increase in stratospheric $CO_2$, one would expect no parameter to change significantly, and thus also the HC to stay rather constant. The biggest uncertainty is in the eddy and the forcing component. GMS and the feedback component should be zero because they both only relate to temperature changes in the troposphere, which we do not expect.

### 2.4.1. Other drivers of HC strength changes

In the previous section we saw three frameworks that could provide physical explanations for HC changes. In the literature, several mechanisms more have been proposed to explain HC weakening with greenhouse gas forcing. There seems to be a 'tug of war' between different mechanisms.

There are two mechanisms relating the weakening of the HC to the direct effect of $CO_2$ increase in the atmosphere. Direct means only considering $CO_2$ effect on radiative transfer without its influence surface temperature changes with their corresponding feedbacks.

Around 81% of the net radiative cooling of the atmosphere is balanced by latent heating which poses a constraint on the precipitation rate (Hartmann, 2016, p. 35). Bony et al. (2013) argue that by increasing the $CO_2$ in the atmosphere, the loss of infrared radiation to space is reduced. This warms and stabilizes the atmosphere weakening upward motions, i.e., the circulation.

The second mechanism employs the spatial structure of $CO_2$ forcing. In regions of ascending motion, such as the equator, the $CO_2$ radiative forcing is reduced. This is due to the presence of deep-convective clouds and the high humidity, which both are effective longwave absorbers. Thus, $CO_2$ below cloud top or the moist atmosphere has a limited effect on OLR (I think this argument is wrong because $CO_2$ band vs. water vapor absorption region). In subsiding regions, such as the subtropics, the $CO_2$ radiative forcing is larger because this effect is missing. Given a larger forcing in the extra-tropics than at the equator, less energy transport and also a weaker circulation is required. About 15% of the century scale weakening is attributed to the direct effect of $CO_2$ using fixed-SST simulations (Merlis, 2015).

Another mechanism that explains the HC weakening compares the change in saturation vapor pressure according to the Clausius-Clapeyron (CC) equation to the changes in precipitation. Assuming that the global mean precipitation is given as

$$P = Mq \tag{2.23}$$

where $M$ is the convective mass flux and $q$ is a typical boundary layer mixing ratio, which scales with the CC equation. In its linearized form, one obtains the following expression for the change in mass flux

$$\frac{\delta M}{M} = \frac{\delta P}{P} - \frac{\delta q}{q}. \tag{2.24}$$

Inserting typical global mean values for $\delta P/P = 0.03\delta T$ (where does this value come from? simulations?) and with CC $\delta q/q = 0.07\delta T$ gives a mass flux change of $\delta M/M = -0.04\delta T$, a weakening of global circulations (Held & Soden, 2006). This relationship holds better for global zonal circulations, e.g., the Walker circulation, than the HC (Vecchi & Soden, 2007).

Similiar to the decomposition of HC changes in the energetic framework, diabatic heating, eddy heat flux, eddy momentum flux, zonal friction and static stability changes can also be seen as the drivers of change in maximum MSF (Vallis et al., 2015). None of these factors

alone can fully explain the changes but the most important contribution according to this decomposition comes from an increase in static stability with diabatic heating changes even contributing to a strengthening of the HC on longer time scales (Chemke & Polvani, 2021). For a comprehensive evaluation and comparison of the different proposed HC weakening mechanisms, one should consult Chemke and Polvani (2021).

Correlating the HC edge with tropical tropopause height and extratropical tropopause height in CMIP models has shown that the HC edge is set by the extratropical tropopause height (Lu et al., 2007) instead of the tropical tropopause height. This favors an eddy driven perspective of the HC edge.

# 3
# Methods

## 3.1. Model setup

### 3.1.1. ECHAM6

How will you be able to reproduce this study? - ECHAM 6 - CMIP 6 type atmospheric GCM - slab ocean ($d$ = 50 m) - resolution: 47 vertical pressure levels with (lat,lon)= (96,192) gridpoints - temporal resolution: ($\sim 200km, 1.875$) seasonal cycle: yes diurnal cycle: yes sea ice: fixed paramterizations: convection, cloud microphysics, radiation, turbulence, land surface, gravity wave drag q-flux correction: yes aerosols: prescribed continental land mass: yes orography: yes insolation configuration: obliquity, eccentricity, precession (?) How do the fixed SST simulations work?

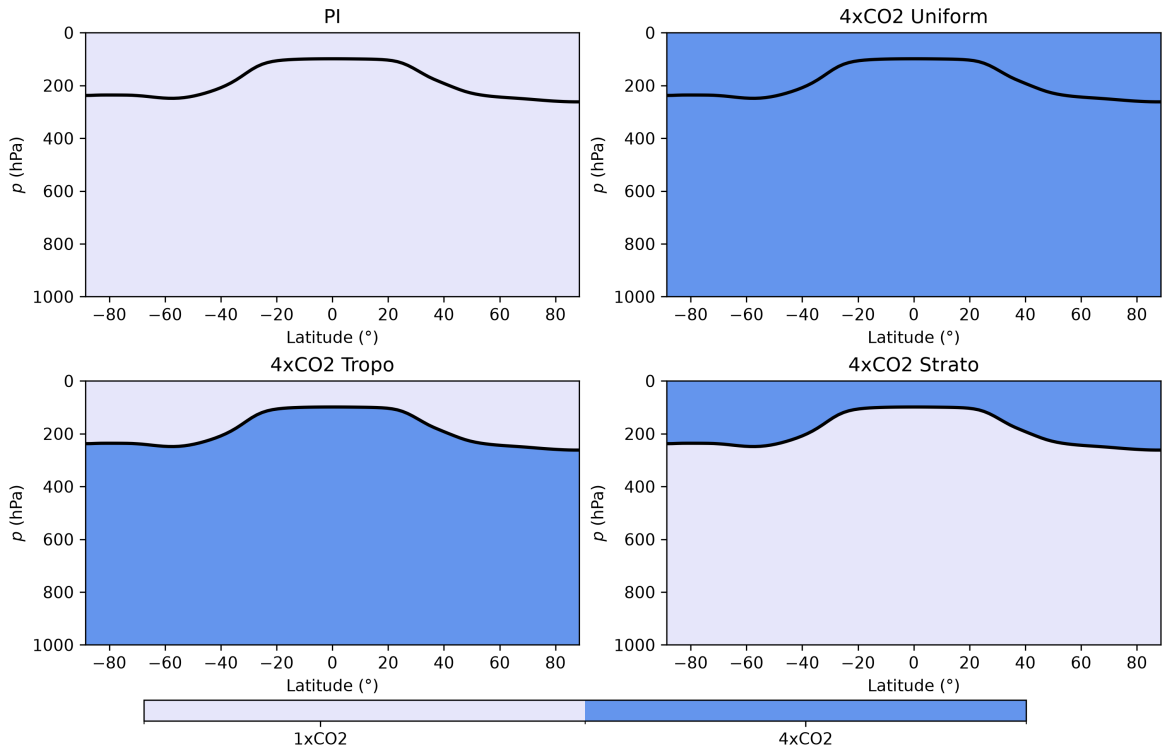



**Figure 3.1**

(TODO: We find it most plausible to calculate the strength of the HC from the maximum of the $\psi_{\max}$ and the extent from the latitude at which $\psi_{\max}$ becomes zero, averaged between 300 and 700 hPa. Thus, all our calculations use these metrics while referenced studies may use different ones.)

### 3.1.2. konrad

- slab ocean - GHG -

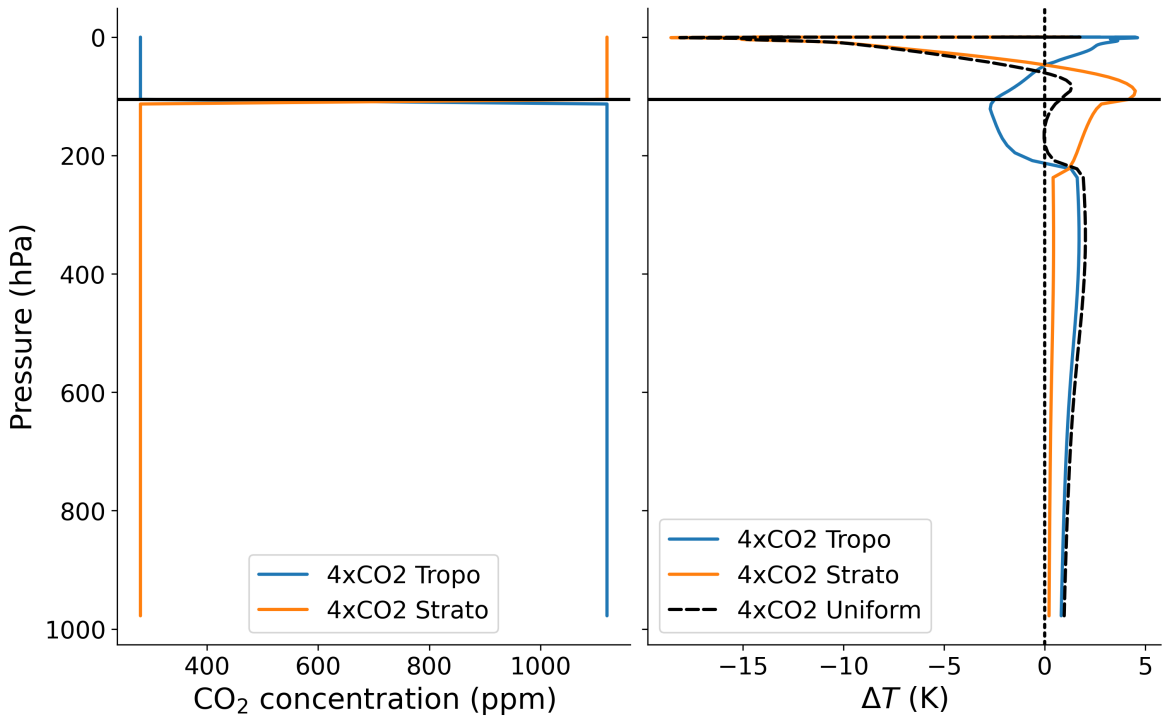



**Figure 3.2**

## 3.2. Definition of the Hadley cell

The HC can be characterized by its strength and extent, both of which can be determined by calculating the meridional mass stream function (MSF)

$$\psi_{\max} = \frac{2\pi a \cos\phi}{g} \int_{p_s}^{0} \bar{v} \mathrm{d}p' \tag{3.1}$$

with the radius of the Earth $a$, the latitude $\phi$, the gravitational acceleration $g$, the surface pressure $p_s$, and the mean meridional wind $\bar{v}$. The MSF is always parallel to the flow in the vertical $p$–$\phi$ plane (see Figure 1.1). A positive value of $\psi_{\max}$ indicates a clockwise circulation and a negative value a counter-clockwise circulation. In the annual mean, the Northern hemisphere shows a clockwise and the Southern hemisphere counter-clockwise circulation. Since the MSF vertically integrates the mass flux, its values symbolize the net magnitude of mass transport above and below and to the right and the left of a certain grid point. At the HC maximum the integral covers exactly the upper or lower branch of the circulation. In the literature, one can find several different metrics for the HC strength. Some use the MSF at a fixed pressure level or spatially average, while others use the minimum of the vertical pressure tendencies $\omega$. HC trends have shown a large sensitivity to the metric used (Pikovnik et al., 2022).

The global HC, as depicted in Figure 1.1, cannot be found at any given longitude but rather there are different regional HCs, which are strongly influenced by the shape and the distribution of continental land mass. HC trends might be different depending on the region

of the HC that is considered. This study will only consider the global HC and, if not stated differently, its annual mean.

## 3.3. From energy imbalance to transport

Integrating this equation in the steady state over a latitude band gives

$$F_{ao}(\phi) = \int_{-\pi/2}^{\phi} \int_{0}^{2\pi} R'_{\mathrm{TOA}} a^2 \cos\phi \mathrm{d}\lambda \mathrm{d}\phi' \tag{3.2}$$

which provides a measure for the energy transport at every latitude. We use the anomalous net radiative TOA flux $R'_{\mathrm{TOA}}$ here, subtracting the global mean value, to make sure that the energy transport is closed (zero transport at both poles) (Zelinka & Hartmann, 2012). To separate between the energy transport by the atmosphere and the ocean, one can subtract the surface energy fluxes, which can be obtained from the sum of surface latent, sensible heat, and net radiative flux, from the TOA radiative fluxes.

## 3.4. Definitions

- tropopause height - HC strength - HC extent - meridional temperature gradient - vertical temperature gradient

# 4

# Results

## 4.1. The atmospheric response

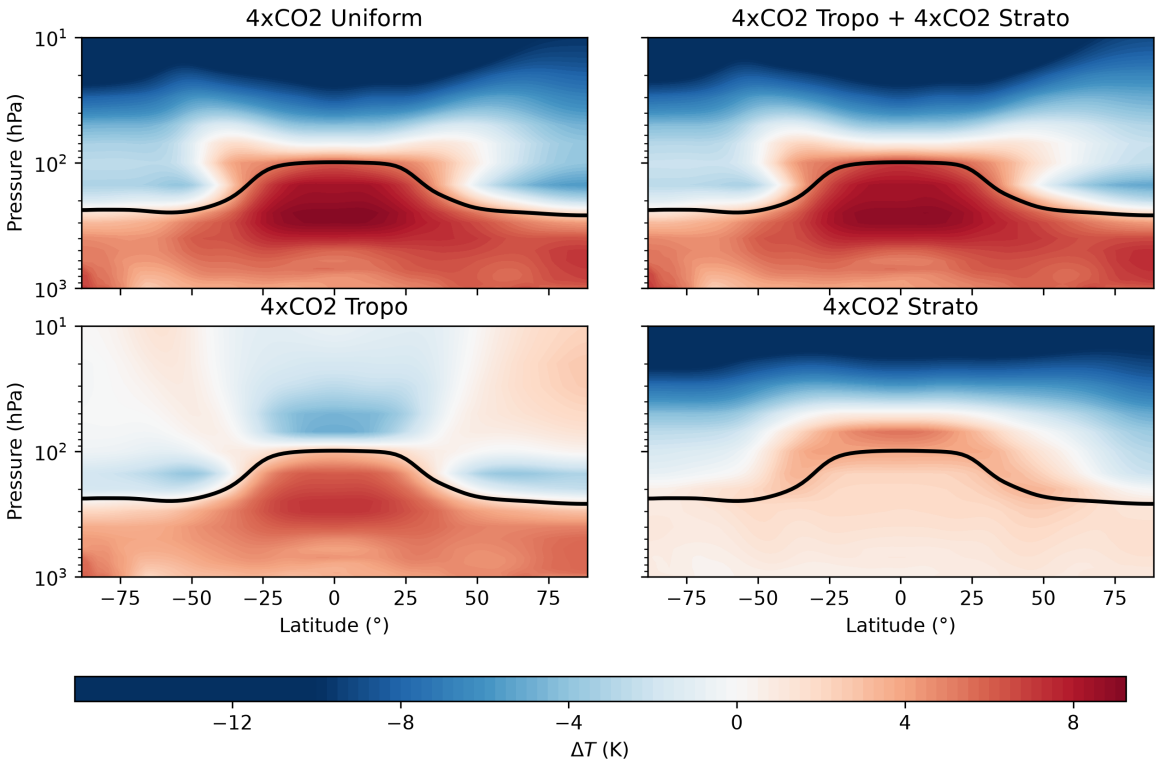



**Figure 4.1**

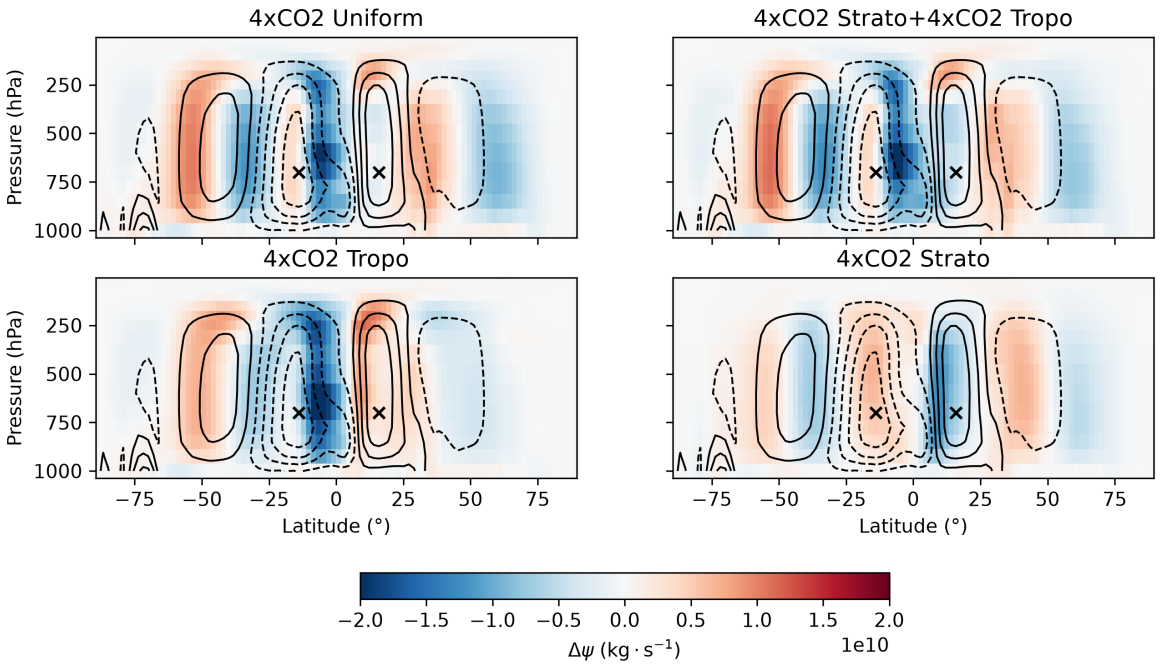



**Figure 4.2**

## 4.2. Predictions from axisymmetric scalings

(Seo et al., 2014)

$$\frac{\delta\psi_{\max}}{\psi_{\max}} = \frac{5}{2}\frac{\delta H}{H} + \frac{5}{2}\frac{\delta\Delta_h}{\Delta_h} - \frac{\delta\Delta_v}{\Delta_v} \tag{4.1}$$

$$\frac{\delta\phi_{\mathrm{H}}}{\phi_{\mathrm{H}}} = \frac{1}{2}\frac{\delta H}{H} + \frac{1}{2}\frac{\delta\Delta_h}{\Delta_h} \tag{4.2}$$

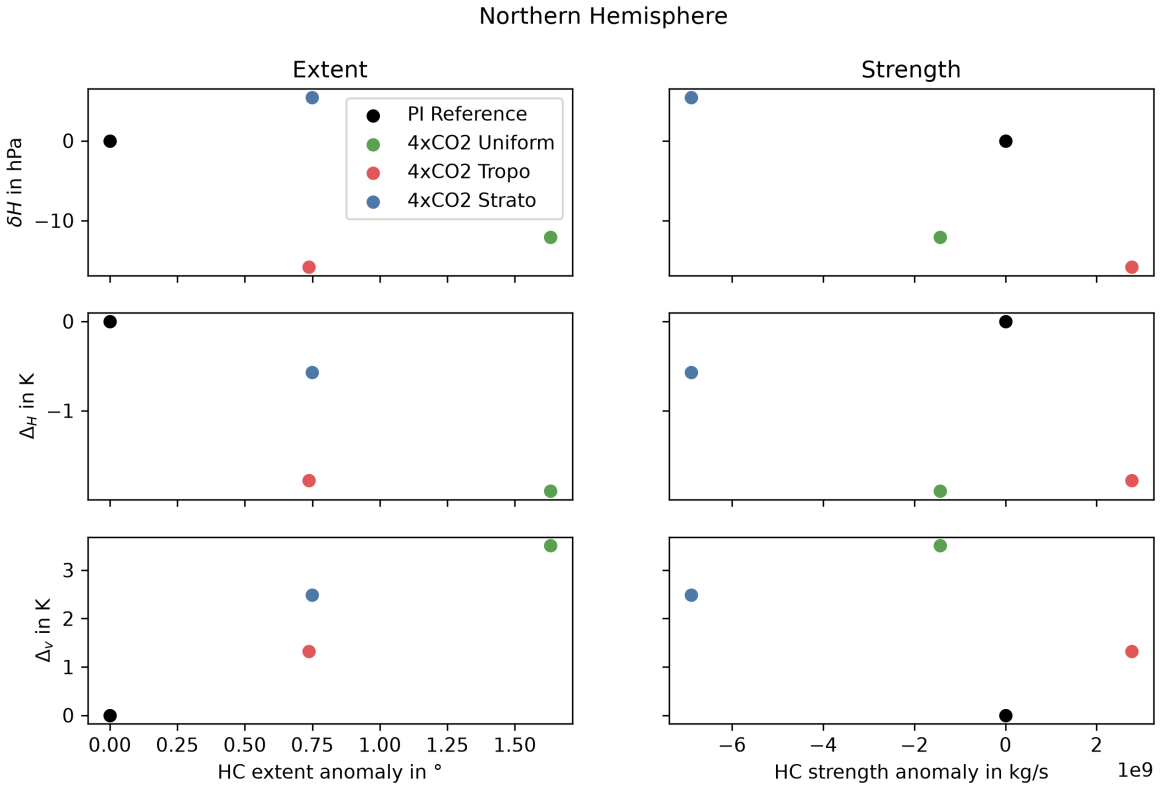



**Figure 4.3**

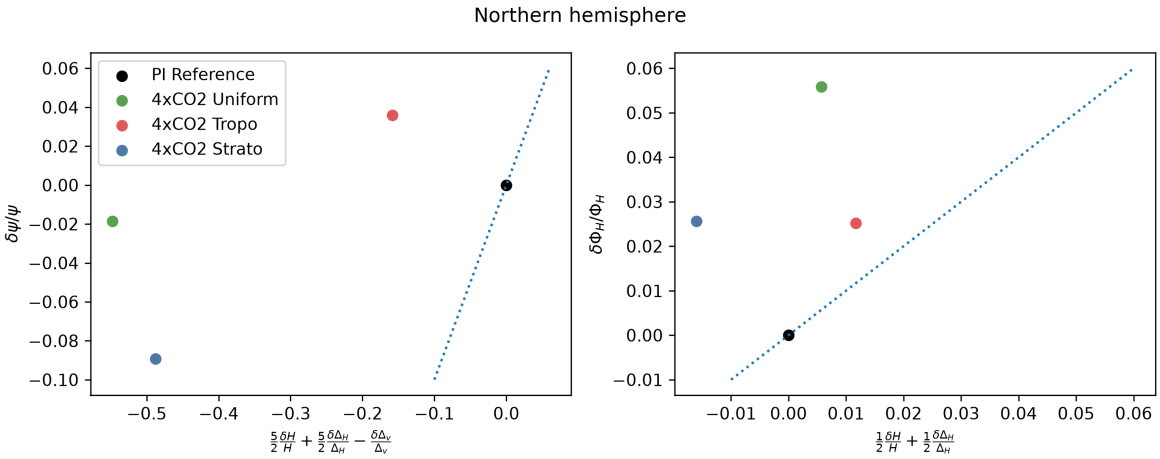



**Figure 4.4**

## 4.3. Eddy driven Hadley cell predictions

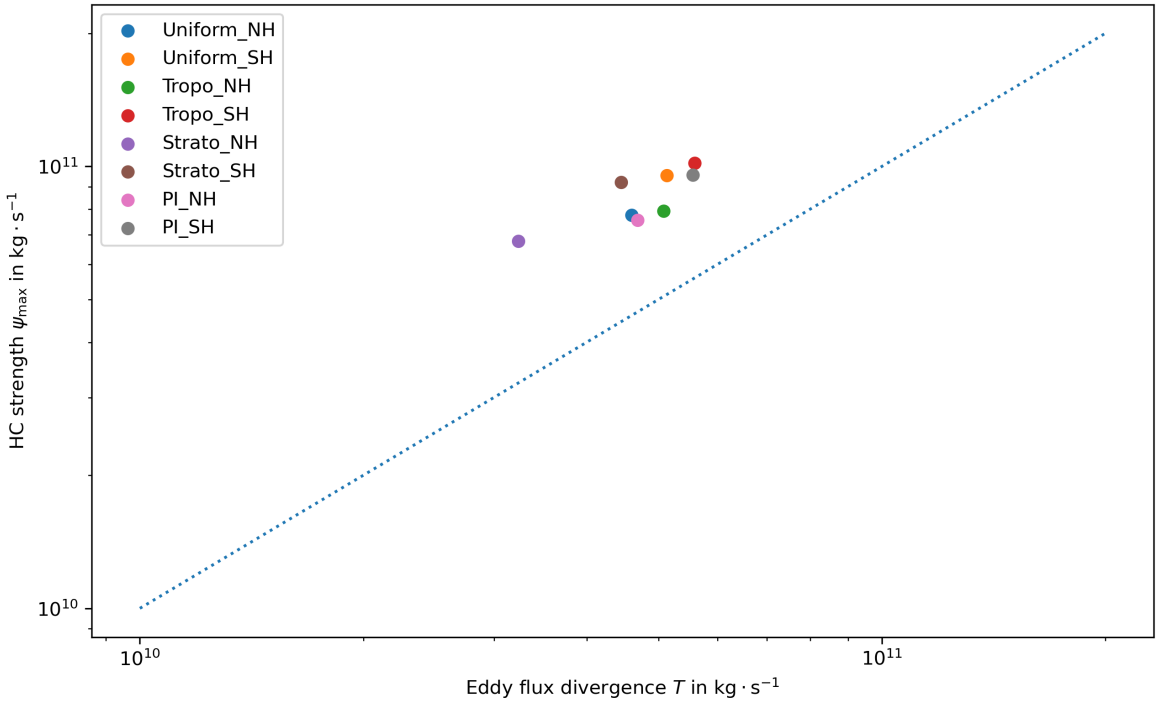



**Figure 4.5**

## 4.4. The energetic framework

### 4.4.1. GMS and energy transport of the Hadley cell

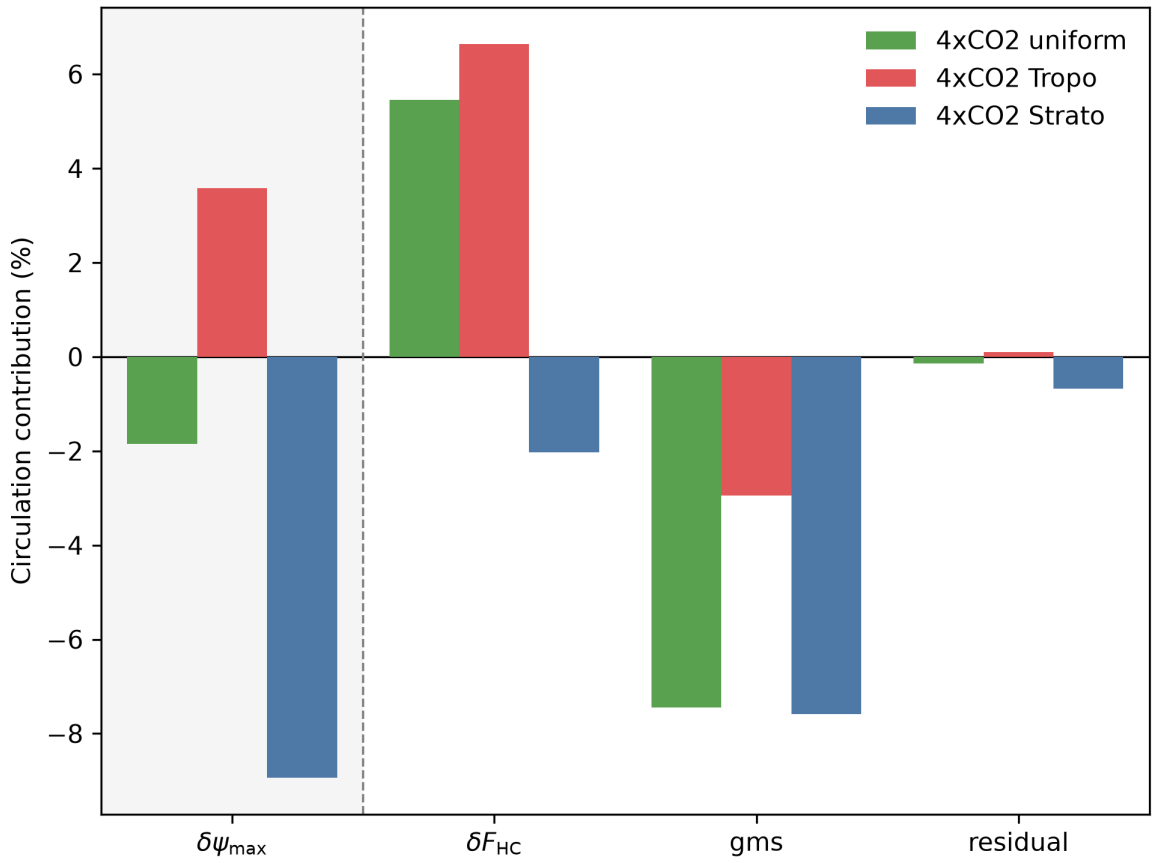



**Figure 4.6**

### 4.4.2. Eddy energy transport

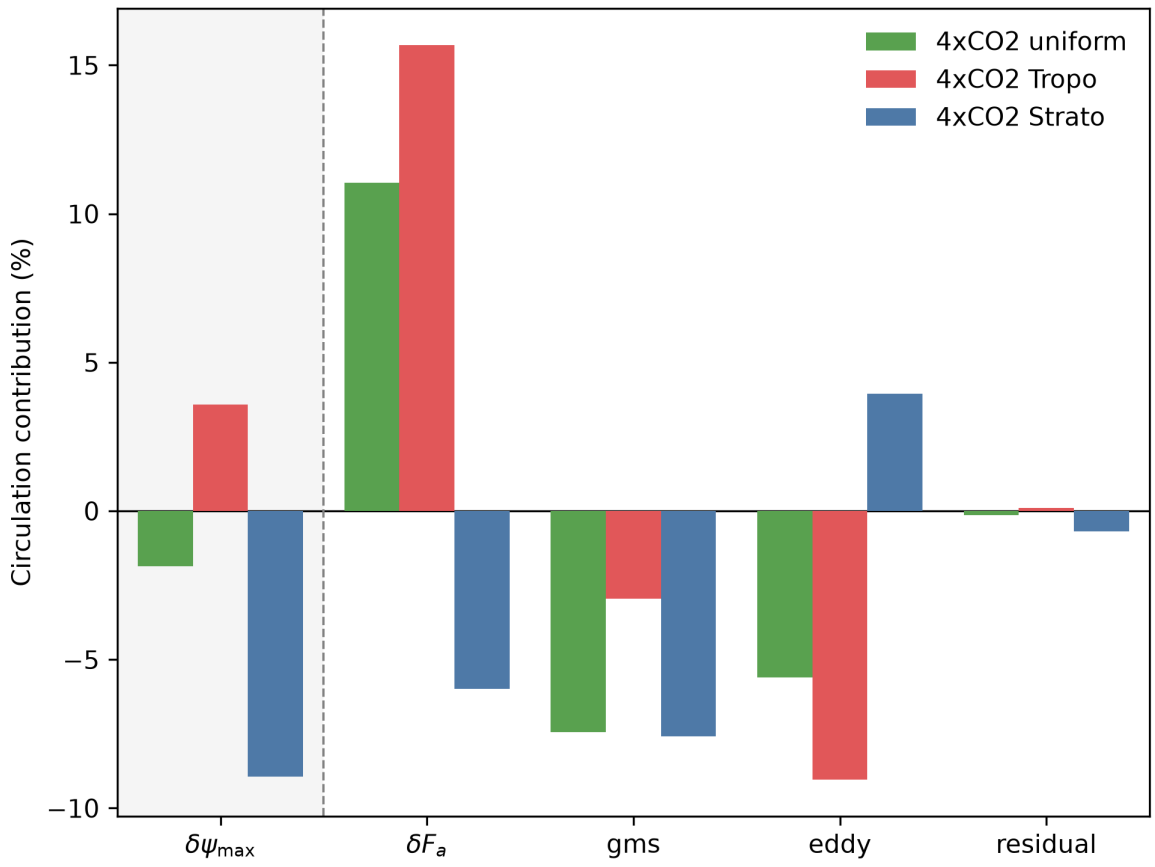



**Figure 4.7**

### 4.4.3. Forcing and feedback contributions

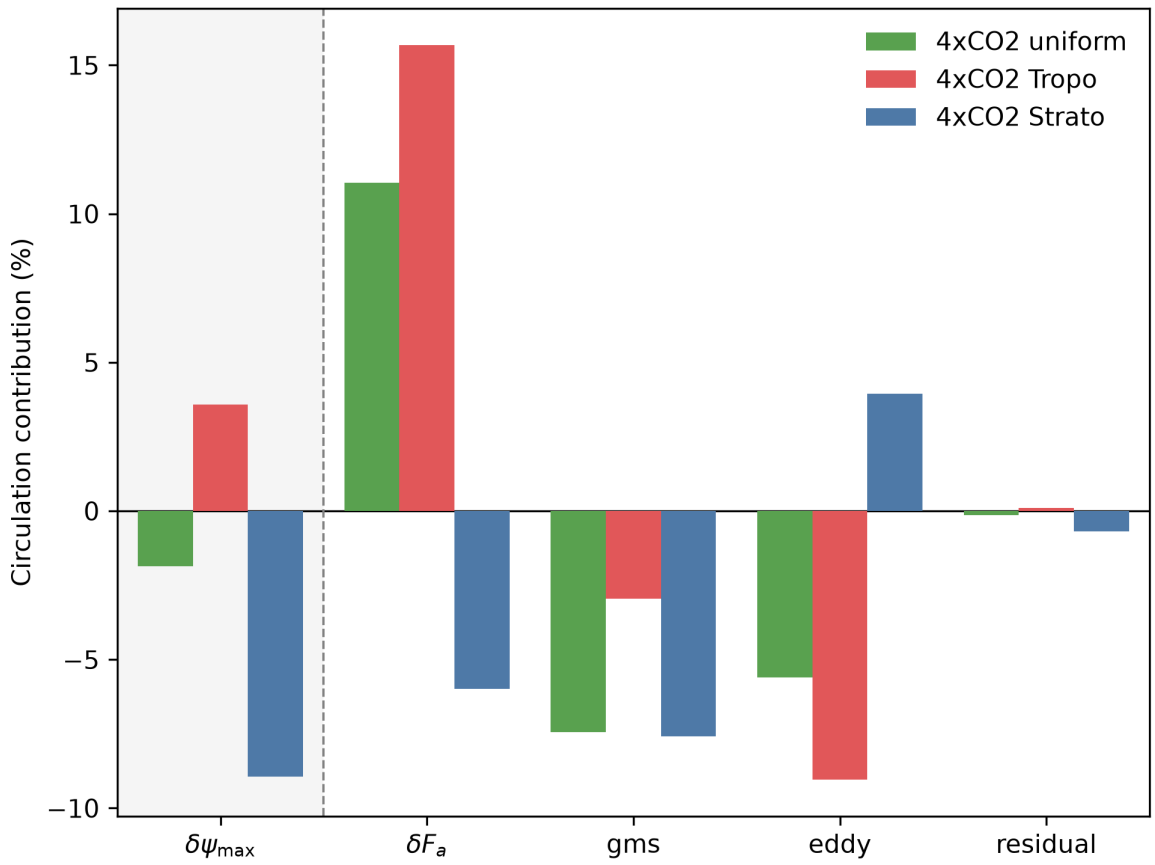



**Figure 4.8**

### 4.4.4. The forcing

Explain the forcing structure (Merlis, 2015)

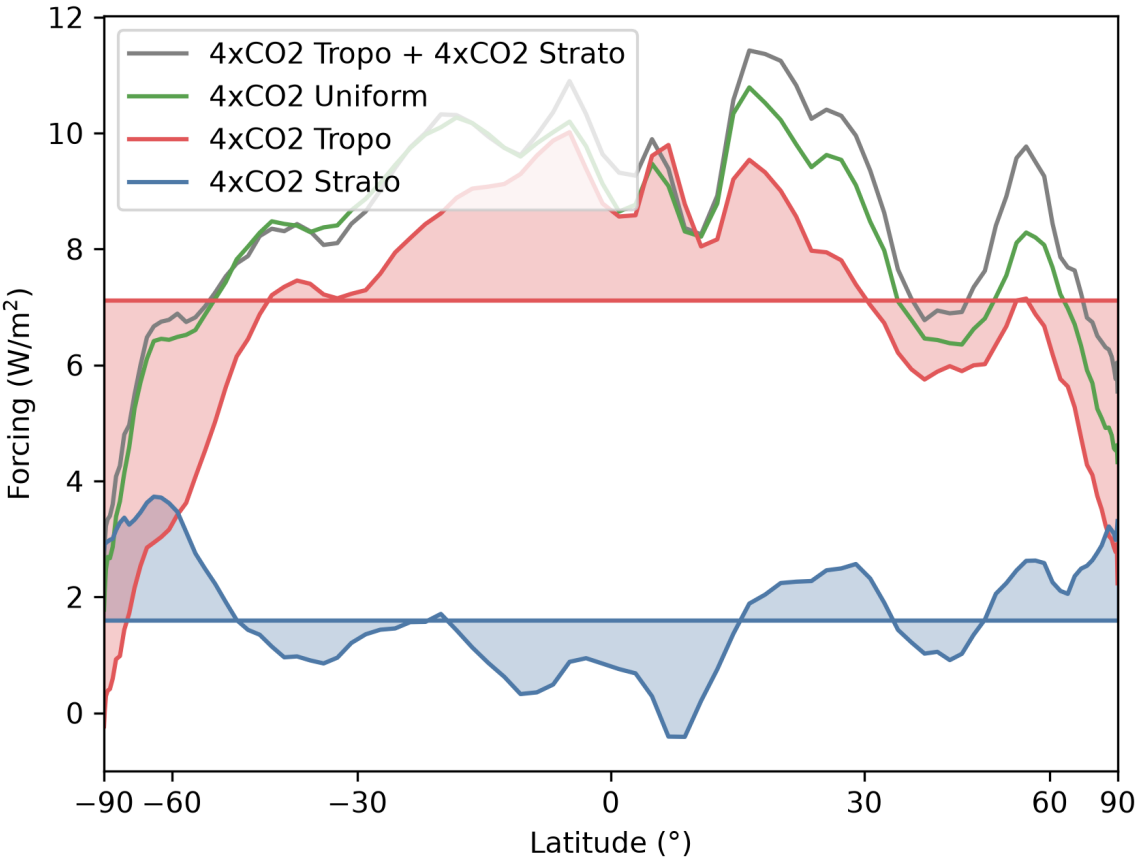

4xCO2 Tropo + 4xCO2 Strato
4xCO2 Uniform
4xCO2 Tropo
4xCO2 Strato
Forcing (W/m²)
Latitude (°)
12
10
8
6
4
2
0
−90
−60
−30
0
30
60
90

**Figure 4.9**

# 5

# Discussion

# 6

# Conclusion

## Acknowledgement

First, I like to thank my first supervisor .... , for .... . Also, I want to thank my second supervisor .... for .... .

Thanks to ..... for all the insights into .... . Thanks to .... for all the intense but fruitful scientific debate about .... . Thanks to ... , who always had an open door for ..... .

Also, I would like to thank the .... .

And last but not least, thanks to .... , who helped ..... . As well as .... for the help with .... .

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